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Organic molecular and polymeric electrets toward soft electronics

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Electrets are a class of dielectric materials with quasi-permanent electric charges or molecular dipoles and the ability to generate electrostatic charge in response to external oscillation and pressure without an external electric field. They can be found in many daily life applications such as microphones and energy harvesters. However, most of the applications developed so far are based on rigid, brittle, solid inorganic, or hard polymeric materials. To develop stretchable/wearable devices, a variety of flexible electret materials is urgently needed. Highly flexible and durable electret materials based on organic molecules and soft polymers are of great interest due to their potential applications in soft electronics, such as pulse wave or heartbeat sensors and haptic control devices. In this mini-review, we briefly present recent developments in organic molecular and polymeric electrets. In particular, we highlight the flexible/stretchable electrets such as the world's first stretchable electret prepared from a molecular liquid. With those recent developments of flexible/stretchable electrets we propose future challenges towards soft electronics.

Design, System, Application

Organic/polymeric soft electrets, which possess flexibility and maintain trapped charges in the materials, are potential materials for soft electronics such as vibration sensors, nanogenerators and flash memories. Most electrets currently used are based on hard and brittle materials that are not suitable for use in the rapidly developing flexible/stretchable electret devices. We focus on several methods and corresponding examples for preparing soft electret materials, particularly an alkyl-porphyrin based liquid. Because the liquid electret has excellent deformability and fluidity, it can be easily embedded into a stretchable device. Improvement of the flexibility and electret performance of polymer-based electrets, including device fabrication methodology, is also necessary in this research field.

1. Introduction

The rapid development expected of flexible electronic devices¹ such as light-emitting diodes (LEDs), e-textiles, sensors, transistors, soft robots, and haptic devices offers great potential for improving everyday life. One of the urgent issues for practical use of such devices connected to the Internet of things (IoT),² such as in healthcare applications, is to develop the flexible/stretchable and durable power source. To solve this issue, organic/polymer electrets are a potential candidate due to their good flexibility, light weight, wear resistance, and large charge storage capacity.

Simply put, an electret is a dielectric material that can store either surface charges, space charges, or dipole charges for long periods of time (Fig. 1A). The word electret first appeared in literature published by British scientist Oliver Heaviside *et al.*³ The term “electret” is composed of “electr” from electrostatics/electricity and “et” from magnets. In 1919, Japanese physicist Mototaro Eguchi used a mixture of natural wax, resin, and bezoar to make the world's first artificial electret through thermal polarisation.^{4,5} Since then, electrets have been systematically investigated by researchers, and various reports^{6–8} have been published. Applications include electrophotography, microphones, air filters, and energy harvesters⁹ (Fig. 1B). The performance of electrets is usually described in terms of two parameters: charge density and charge life. Charge density is a measurement of the storage capacity of the space charge or polarisation charge of the electret, including the surface charge density and space charge density. It is generally represented by the symbol σ , in units of $C\ m^{-2}$ or $C\ m^{-3}$. The quantitative description of the electret charge stability, that is, the effective decay time

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constant of the electret charge, is also called the electret charge life. For designing a stretchable electret, the basic principle is to increase the charge density and charge life as much as possible on the premise of achieving stretchability.

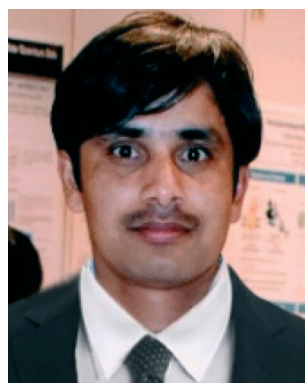
Crucial factors for electret performance besides charge density and charge life are the composition and orientation of the dielectric materials. Therefore, the types of electrets can be classified in terms of dielectric materials and their carrier types. Dielectric materials can be separated into inorganic and organic materials. The development of inorganic materials is mainly concentrated on silicon dioxide (SiO_2),^{10–12} barium titanate (BaTiO_3),¹³ aluminium oxide (Al_2O_3),¹⁴ lead zirconate titanate (PZT),¹⁵ zinc oxide (ZnO)¹⁶ and so on. Even though this type of electret exhibits high surface charge density, it suffers from poor durability and

unreliability.¹⁷ While among organic electrets, the dominant dielectric materials are polymers, such as polytetrafluoroethylene (PTFE),¹⁸ polyethylene (PE),^{19–21} polypropylene (PP),²² polyethylene terephthalate (PET),²³ polyimide (PI),²⁴ poly(methyl methacrylate) (PMMA),²⁵ polyvinylidene fluoride (PVDF),²⁶ and ethylene vinyl acetate (EVA).²⁷ Typical representatives of organic electret are shown in Table 1. They all show high insulating properties. Recently, small molecules have also been utilised to form electret materials. The details are described in sections 3.3 and 4.5. We compare inorganic and organic electret about represented materials,²⁸ surface density of the charge,^{11,29} substrate,³⁰ charge life,^{9,12,31} hardness, thickness,^{29,32} processability, and applications³³ (Table 2). It is worth mentioning that inorganic electrets usually exhibit a higher σ than that of



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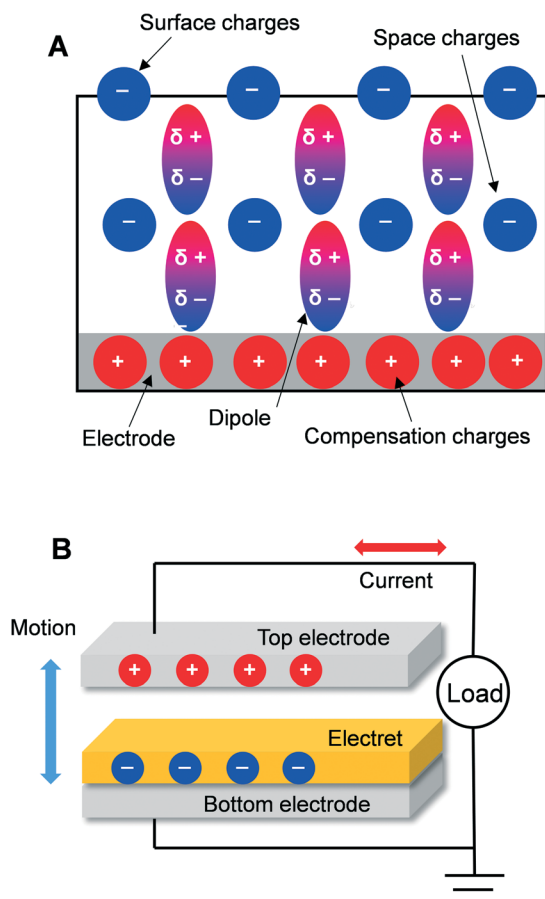


Fig. 1 (A) Schematic diagram of the charge distribution in an electret material. (B) A typical power generation mechanism at a simple electret electrode-device.

organic electrets with a shorter charge life. Unlike inorganic electrets, organic ones show adaptability and compatibility with substrates towards soft electronics because of their

flexibility and processability. Inorganic electrets are often used in micromachining technologies, sensors, microphones, and transducers, while organic electrets are developed in the fields of microphones, transducers, memory devices, energy harvester, and healthcare devices.³³ Electret charges may store real charges, dipoles, or both, with these being stored either on the surface of the electret or in spaces throughout the material (Fig. 1A).³⁴ Depending on the type of charge carrier, electrets are divided into real-charge electrets and orientated-dipole electrets. Real-charge electrets include surface charge that appears on the surface of the dielectric material, and space charge that presents in the bulk volume of the dielectric material. Orientated-dipole electrets contain orientated (aligned) dipoles. This type of charge is formed by freezing the orientation dipole and interface polarisation.

In this mini-review, we briefly introduce the common charging methods, focusing on the development of organic molecular and polymeric electrets in recent years. We then focus on the flexibility or stretchability of electrets, including the world's first stretchable "liquid electret" prepared from room-temperature liquid molecules. Lastly, we discuss the challenges and prospects associated with the development of more efficient organic molecular and polymeric electrets toward soft electronics.

2. Charging methods

There are many ways to generate electrets, and various charging methods exist to suit the different types of electret carriers (real charge or dipole charge). Almost all real-charge electrets are obtained by discharging methods, such as corona charging,^{35–37} electron beam irradiation,^{38,39} and liquid contact charging.^{36,39,40} Oriented-dipole electrets are formed by applying an electric field at room temperature to



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Table 1 The name and chemical structure of some polymer electret materials

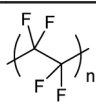
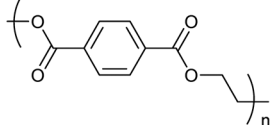
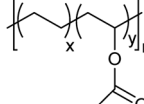
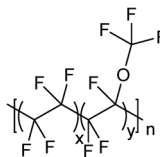
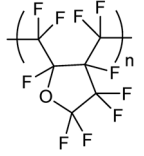
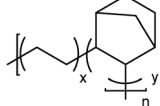
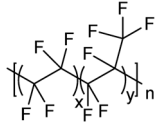
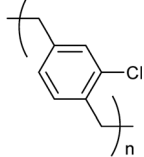
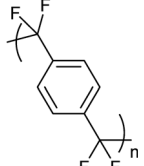
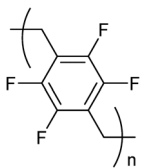
Name	Abbreviation	Chemical structure	Ref.
Polytetrafluoroethylene	PTFE		18
Polyethylene terephthalate	PET		23
Poly(methyl methacrylate)	PMMA		25
Poly(vinylidene fluoride)	PVDF		26
Ethylene vinyl acetate	EVA		27
Perfluoroalkoxy alkane	PFA		48–50
Poly(perfluoro-3-butenyl-vinyl ether)	CYTOP		35, 51
Copolymers of cycloolefin	COC		57
Fluorinated ethylene propylene	FEP		78
Parylene C	—		31
Parylene AF-4	—		29, 31
Parylene VT-4	—		31

Table 2 Comparison between inorganic and organic electrets

Property	Inorganic electrets	Ref.	Organic electrets	Ref.
Materials	Metals, metal oxides, ceramics	28	Small molecules, polymers	28
Surface charge density (σ)	$<8.7 \times 10^{-3} \text{ C m}^{-2}$	11	$<3.7 \times 10^{-3} \text{ C m}^{-2}$	29
Compatibility with substrates	Poor	—	Good	30
Charge life	>2 years	12	Polymer film; >280 days	31
			Stretchable electret; >60 days	9
Hardness-flexibility	Less flexible	—	Flexible	—
Thickness	$<1 \mu\text{m}$	32	10–50 μm	29, 32
Processability	Unshapable	—	Shapable	—
Applications	Micromachining technologies, sensor, microphones, transducers	33	Microphones, transducers, memory devices, energy harvester, healthcare devices	33

polarise the dipole inside the dielectric material. Some typical charging methods are explained as follows.

2.1 Corona charging

Corona charging, the main principle of which is point discharge, is a common methodology found throughout the

literature related to electret study.⁴¹ A typical corona charging device is shown in Fig. 2A.^{35,37} When a high direct current (DC) voltage of thousands of volts is applied between the needle and the ground plate electrode, the air between the two will be broken down, causing a corona discharge phenomenon.³⁷ Charges with the same polarity as the needle tip will accelerate to the ground plate electrode under the

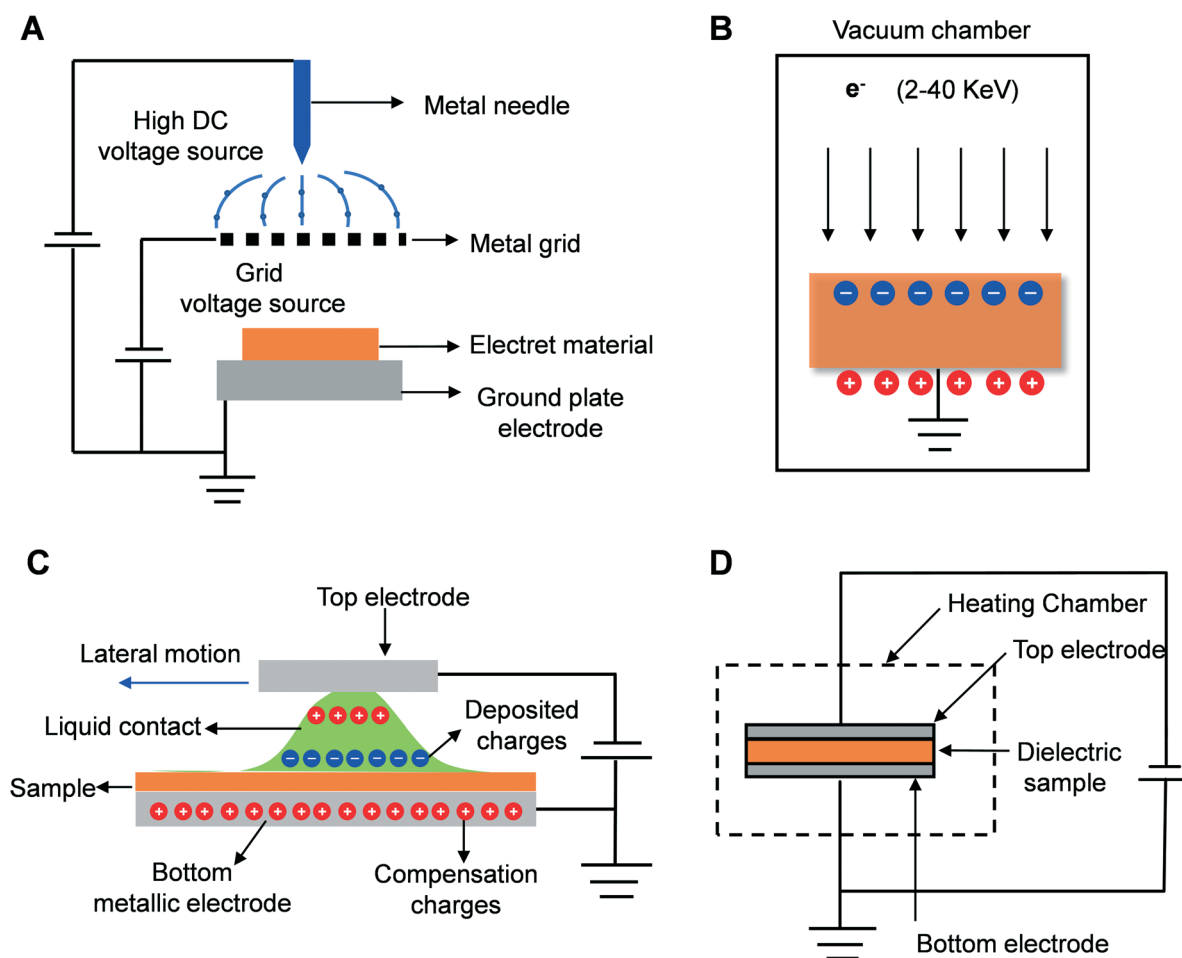


Fig. 2 Schematic diagrams of (A) typical corona charging, (B) experimental setup for forming thin-film electret by monoenergetic electron scanning charging, (C) the device for charging the sample by the liquid contact method, and (D) electrothermal polarisation device.

action of the electric field. At this moment, the material attached to the plate electrode will trap the charge and become an electret. The electric field intensity between the needle tip and the plate electrode is not uniform. To obtain an electret with a homogeneous potential distribution, a metal grid is usually placed between the needle and the material. It is necessary to provide voltages of the same polarity on the power grid to form a grid-controlled constant voltage corona charge, so that the lateral distribution of the captured charges of the sample is uniform.

2.2 Electron beam irradiation charging

Electron beam irradiation charging generally refers to non-penetrating monoenergetic electron beam irradiation charging.³⁹ Fig. 2B depicts a schematic diagram of modern electron beam irradiation charging.³² The charging process is generally under vacuum conditions, using a low-energy electron beam ($\sim 10\text{--}50$ keV) to irradiate the sample. The range of penetrating electrons is controlled and kept smaller than that of the thickness of the dielectric material. Once the electrons are injected into the dielectric material, they are quickly trapped inside and produce negatively trapped space charges. A common charging model is a single-sided, vapour-deposited aluminium electrode film. The beam current density is usually in the order of 10^{-4} A m $^{-2}$, the irradiation time is a few seconds to tens of seconds,³⁸ and the thickness of the dielectric sample film is $1\text{--}50$ μm . Compared with other charging methods, the advantage of electron beam irradiation is that the distribution and changes of the space charge can be studied under controlled conditions. The charge density can also be accurately controlled by adjusting the energy of the electron beam and injected beam current.

2.3 Liquid contact charging

Liquid contact charging is an approach using charge transfer through a liquid.⁴⁰ As displayed in Fig. 2C, two electrodes are used on the top and bottom sides of the dielectric material. The bottom electrode is a simple metallic plate and the top electrode is made up of fabric wetted with conductive liquid.³⁶ Liquids suitable for contact charging include ethanol, acetone, dilute hydrochloric acid solution, dilute sodium hydroxide solution, and so forth. When the electric field is applied between both electrodes, the interaction of the electrostatic and intermolecular force will occur from the top electrode toward the dielectric material surface.⁴² In this method, the top electrode can be moved over the surface of the material, enabling charge transfer over a large area of the material.⁴³ A set time is necessary for the transfer of sufficient charges, and after that time, the moveable top electrode is shifted to the next position. The charge density on the dielectric material surface depends on the charging time as well as the applied voltage. To ensure that the charges can be stored in the dielectric material for a long time, the electrode must be withdrawn and the contact liquid should be evaporated before removing the potential.

2.4 Electrothermal polarisation charging

As Fig. 2D illustrates, the electrothermal polarisation approach⁴⁴ applies an external electric field to the dielectric material at a specified high temperature, so that the molecular dipoles are orientated along the direction of the electric field. After that, the environment is cooled to a lower temperature while maintaining the electric field to freeze the dipoles.⁴⁵ In this method, due to the different contact conditions between the external electrode and the dielectric sample, the air gap between the electrode and the sample may produce complicated charging phenomena. There are three such phenomena: i) the orientation of the dipole causes internal polarisation of the hetero-charges; ii) the spark discharge between the air gaps leads to the deposition of the homo-charges; and iii) homo-charges are injected through the contact electrode.^{33,46} The electrets formed by thermal polarisation are called thermal electrets.⁴⁷

3. Development of organic electrets

In the past few decades, researchers have focused more on polymers for organic substances of electrets because they have a number of advantages including better thermal stability, good mechanical properties (such as high strength and creep resistance), and relatively easy processing.

3.1 Fluoropolymers

Among the common polymers, fluoropolymers are an important organic electret material. This type of material has good dielectric properties and excellent chemical inertness. It can be used in harsh environments such as high or low temperature, and humid conditions. Charges, especially space charges, injected into the fluoropolymer will be isolated. Therefore, its ability to store charges is excellent. Since fluorine is the most active non-metallic element in nature, the bond with carbon atoms is sufficiently strong, which is the reason for the chemical inertness of fluoropolymers.³² Representative fluoropolymers are PTFE,¹⁸ perfluoroalkoxy alkane (PFA),^{48–50} and poly(perfluoro-3-butenyl-vinyl ether; cyclic transparent optical polymer; CYTOP^{35,51} (see Table 1).

a. PTFE. PTFE possesses excellent chemical stability and dielectric properties and shows outstanding performance as a real-charge electret material. In addition, it has extremely low water absorption and good high-temperature resistance; thus it exhibits excellent space charge storage capabilities.⁴⁹ Wang *et al.*⁵² obtained a value of σ as high as 1.0×10^{-3} C m $^{-2}$ by integrating a triboelectric nanogenerator (TENG) with PTFE, which is 20 times higher than that of traditional TENG samples. The porous morphology of PTFE materials prepared by corona charging provides an outstanding charge storage capacity at high temperature compared with other fluoropolymers.^{53,54} For example, Xia *et al.*⁸ kept charged samples of porous PTFE at 300 $^{\circ}\text{C}$ for 5 h, after which the equivalent surface potential (V_s) still retained more than 90%

of the initial value, while the V_s of ordinary PTFE decayed to 40% of the initial value. However, a fatal disadvantage of PTFE is its poor processability. It will exhibit plastic deformation (creep) under long-term continuous load, which brings certain limitations to its application.

b. PFA. PFA is similar in nature to PTFE. However, its additional $-\text{OCF}_3$ group gives it improved high-temperature characteristics over PTFE. For example, it is still flexible at 160 °C and can work continuously for 104 h at 285 °C.^{48–50} The main drawback of PFA is its higher cost compared to other fluoropolymers.

c. CYTOP. Good processability originating from high solubility in organic solvents, amorphousness, and transparency are specific characteristics of CYTOP. For traditional electrets, most of the materials are insoluble in solvents and coloured; thus their practical usability is restricted. As a promising electret material, Suzuki *et al.*^{35,51} found CYTOP, which had a dielectric strength value of up to 110 kV mm⁻¹. CYTOP can be easily fabricated into films, and it has a high compatibility in microelectromechanical systems. Sakane *et al.*³⁵ showed that the σ of CYTOP electrets improves significantly with the addition of the amidosilyl end group. A σ of 1.5×10^{-3} C m⁻² was obtained for a 15 μm thick film, which remained constant for over 4000 hours. The electret film exhibited a micro power generation performance of 0.7 mW at a low seismic frequency of 20 Hz, which is about 2.5 times higher than an earlier result.⁴⁵ Kashiwagi *et al.*⁵⁵ introduced aminosilane derivatives based on CYTOP to form nanoclusters combined with the polymer matrix in the electret film. Nanoclusters act as charge trap sites and enhance the surface charge density and thermal stability of the trapped charges. Under the optimum corona charging conditions, the 15 μm thick electret film achieved an extremely high surface charge potential of -1.6 kV and σ of 2.0×10^{-3} C m⁻².

Zhang *et al.*⁵⁶ used a deep-learning-based *de novo* molecule generator to screen for a high-performance polymer electret material among many candidates. They found that the hydroxyl group was critical for increasing the electron gain energy, and so they designed a CYTOP derivative with *N*-(3-aminopropyl)diethanolamine. The film with a thickness of 15 μm retained a surface potential of -3.0 kV for more than 800 h, far exceeding that of other commercial CYTOP polymer electrets.

3.2 Non-fluoropolymers

Although fluoropolymers play an important role in polymeric electrets, non-fluoropolymers also have potential as good electret materials. Such non-fluoropolymers include polyimides (PI) and copolymers of cycloolefin (COC)⁵⁷ (Table 1). These polymers could be applicable in many fields such as xerography⁵⁸ and air filters.⁵⁹

The main chain of PI contains repeating imide groups with cyclic structures; therefore, it has good flexibility and relatively high thermal stability with a melting point over 500

°C.⁶⁰ PI-based electrets are widely used in resistor-type memory devices^{61,62} and provide advantageous properties such as thermal stability, chemical resistance, and homogeneity to these devices.

Low density, high transparency, low moisture absorption, and excellent mechanical properties⁶³ are the typical characteristics of COC. Because its structural composition contains only non-polar C–C bonds, COC exhibits low dielectric properties. Camacho *et al.*⁵⁷ found that the surface of the electret of COC derivatives often contained a high concentration of trapped charges, which can maintain a high potential level at room temperature for a long period of time.

3.3 Small molecules

In the past, most of the research on electrets has concentrated on polymers. However, in recent years, scientists have paid more attention to small molecules due to their benefits such as limitless selection, easy preparation, and softness with lower moduli compared to polymers. Therefore, related research studies and reports are gradually increasing. For instance, Tanaka *et al.*⁶⁴ recently developed a self-assembled electret based on 1,3,5-tris(1-phenyl-1*H*-benzimidazol-2-yl)benzene (1). They demonstrated a vibration-based electret based on the self-assembled film of 1 (Fig. 3). This can be an electret generator (EG) that does not require any charging processes. The EG can provide a relatively high

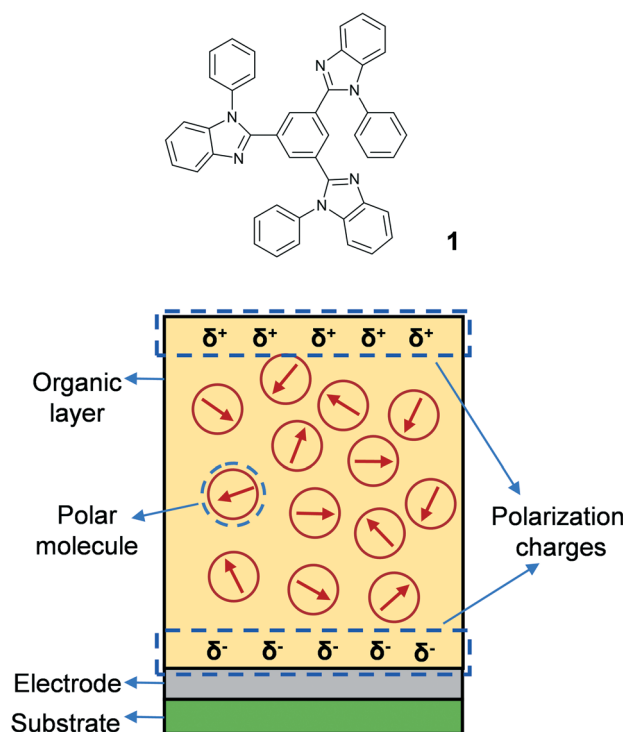


Fig. 3 Chemical structure and schematic illustration of organic thin film composed of polar molecule with spontaneous orientation polarisation. Adapted from ref. 64 with permission from the © 2020 under a Creative Commons Attribution 4.0 Unported (CCBY) license.

output voltage even at low vibration frequencies without using an external bias source. The σ is higher than that of a previously reported system based on bathocuproine (BCP).⁶⁵ The self-assembled film of **1** exhibited a σ of $1.7 \times 10^{-3} \text{ C m}^{-2}$ without a charging process, which is equivalent to the traditional polymer electrets after charging. A self-assembled electret enables the integration of an electret with a micro-electro-mechanical system (MEMS) vibration device by evaporation as a post-process.⁶⁶ Due to the spontaneous orientation of polar molecules, the surface potential ($1.7 \times 10^{-3} \text{ C m}^{-2}$, around 60 V) of self-assembled electret can be established in the microscopic region ($20 \times 20 \mu\text{m}^2$) at room temperature without the charging process.

4. Soft/stretchable electrets

A major issue of electrets is the trade-off between the longevity and stretchability. For today's practical applications, inorganic electrets still occupy the majority.⁹ Even the inorganic electrets (hard electrets) trap charges/dipoles efficiently but lack stretchability. On the other hand, soft organic electrets like rubber bear sufficient stretchability but they discharge quickly, *i.e.* they are short-lived. The solution to this conflict is to make a soft electret from a hard one using different ways. There are also a number of challenges applied from molecular/polymeric materials. Recent examples are presented as follows.

4.1 Introduction of pores in hard electrets

Inorganic and hard plastics possessing pores inside can be stretchable electrets.⁹ Wegener *et al.*⁶⁷ found that the optimal charging of porous PTFE film electrets depended on changes in the charging parameters, *e.g.* the electric field and charging time. Also, the proper charging will result in high surface potentials observed on unstretched or slightly stretched porous PTFE membranes. However, these stretches are small and easily collapsed, because of the lack of ductility of PTFE and the air space in the pores.^{68–70}

4.2 Elastomer with nano-fillers

Carbon nanotubes are introduced into an elastomer matrix to enable stretchable electrets.^{9,71,72} The poled composite discharges through the elastomer within a few hours. To attain a longer charge life, nanoparticle–elastomer composites were prepared. This is a widely used method with many published reports.^{9,73,74} For instance, Suo *et al.*⁹ immobilised nanoparticles in the matrix of an elastomer. After poling, the interfaces between the nanoparticles and the matrix trap charges. Charges of certain amounts and different polarities may be trapped in the matrix. Charges are also trapped in the elastomer but they are short-lived. Similar to the other electrets, the composite electret can be used with no electrode, one electrode, or multiple attached electrodes. Suo *et al.*⁹ reported a polydimethylsiloxane (PDMS) elastomer with silica nanoparticles, which provided a charge

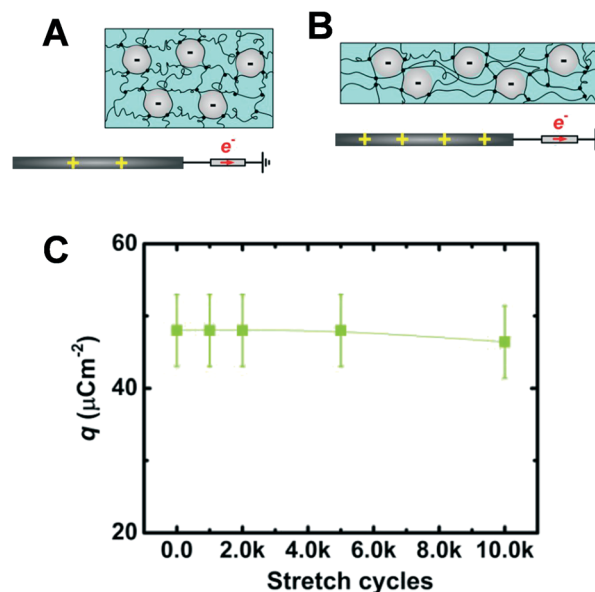


Fig. 4 (A) A matrix of a dielectric elastomer immobilises silica nanoparticles of a hard electret. Upon poling, the electret particles trap charges. A rigid electrode is connected through a resistor to the ground. When the composite approaches the electrode, charges are induced on the electrode, drawn from the ground. (B) When the composite is stretched, the deformation of the elastomer carries the electret particles, changes the amount of induced charges on the electrode, and causes a current through the resistor. (C) Charge stability of PDMS with 1.0% weight ratio of fillers after cyclic stretch. Adapted with permission from ref. 9, copyright 2020 American Chemical Society.

density of $\sim 4 \times 10^{-5} \text{ C m}^{-2}$ and a charge life beyond 60 days. As shown in Fig. 4A and B, the principle of the retractable electret is depicted in detail. At first, the hard electret silica nanoparticles were immobilised in a matrix of PDMS (Fig. 4A). After polarisation, the particles and the interface between the nanoparticles and the matrix trap charges. When the composite electret is stretched, the deformation of the elastomer carries the electret particles, changes the amount of induced charge on the electrodes, and causes current to flow through the resistor (Fig. 4B). Both rigid body motion and deformation of the composite electret generate charges on the electrodes. In particular, the authors performed cyclically stretched electret experiments with an amplitude $\lambda = 1.2$ and a frequency of 2 Hz. After 104 stretching cycles, the charge density remained at 85% of the initial polarisation value (Fig. 4C).

4.3 Preparation of flexible thin sheets

The use of flexible thin sheets is a common method used to prepare flexible electrets.^{75–78} Ma *et al.*⁷⁸ prepared a FEP film electret with wave-shaped structure through template patterning method. The wave-shaped electret was applied to an electrostatic energy harvester. Its surface potential of -500 V and a seismic mass of $3g$, where g is the acceleration due to gravity, were obtained. The energy harvester generated by

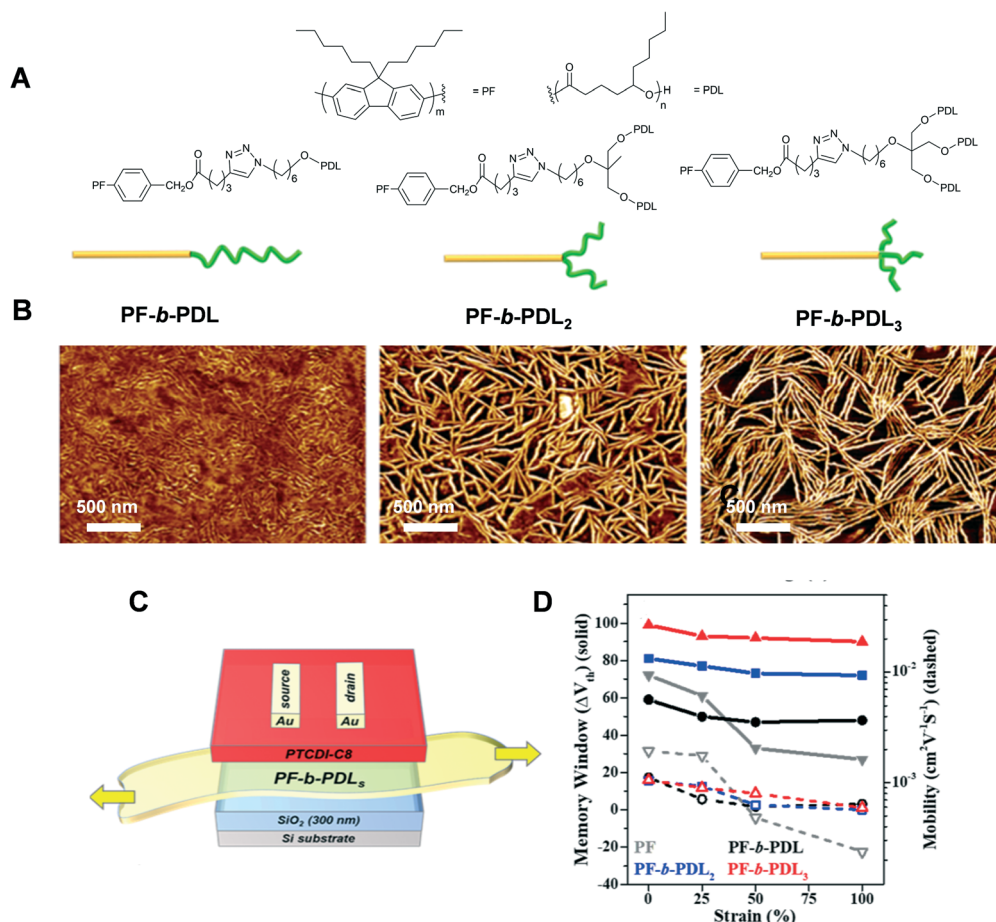


Fig. 5 (A) Chemical structure and model of block copolymers PF-*b*-PDL, PF-*b*-PDL₂ and PF-*b*-PDL₃. (B) AFM phase images of block copolymers PF-*b*-PDL, PF-*b*-PDL₂ and PF-*b*-PDL₃. (C) Structure of the stretchable OFET memories. OFET memories were fabricated with different polymer electrets (Si/SiO₂ (300 nm)/PF or PF-*b*-PDL_x/PTCDI-C8/Au). (D) Memory window (solid lines) and charge mobility (dashed lines) of the stretchable memory device under different strain (%). Reproduced from ref. 79 with permission from the copyright © 2021, American Chemical Society.

the optimal load resistance is 355 μ W at 22 Hz, under 1g of electrostatic energy input acceleration. Tensile studies showed that the Young's moduli of the flat FEP and the wavy FEP were 272 and 3 MPa at strains between 2% and 4%, respectively. This reduced Young's modulus of the wavy FEP film is beneficial for energy harvesting applications as it is more easily deformed under ambient vibrations.

4.4 Use of block copolymers

Block polymers are another candidate for preparing flexible/stretchable electrets.^{74,79,80} Chen *et al.*⁷⁹ designed and synthesised a series of block copolymers based on polyfluorene (PF) and poly(δ -decanolactone) (PDL): PF-*b*-PDL, PF-*b*-PDL₂, and PF-*b*-PDL₃ (Fig. 5A). Among these, the fibrous PF nanostructure serves as an effective hole trapping site and can achieve controllable memory performance, while PDL is a bio-based amorphous polymer with a glass transition temperature (T_g) below -60 °C, which is a key part of block copolymer softening. Soft segments with different structures not only provide elasticity but also adjust the phase separation and charge trapping sites in the polymer film (Fig. 5B and C). The organic field-effect transistor (OFET)

memory device with PF-*b*-PDL₃ as the electret layer exhibits the largest memory window† (102 V) and can maintain its performance under strain up to 100% (Fig. 5D).

4.5 Functional molecular liquids

In most of the above-described electret materials, the charges are trapped at the surface of the electret film; hence the total charge density is limited. To overcome this issue along with stretchability, our group made an electret based on a functional molecular liquid (FML) 2 (Fig. 6A and B), in particular, a liquid porphyrin.⁸¹ FMLs^{82,83} are of special interest because of their reliable properties independent from the device shape and usage environment, even under highly deformable conditions. FMLs exhibit the unique properties of the liquid physical state of the materials under solvent-free

† Memory windows: for OFET memory, the memory window is defined as the threshold voltage (V_{th}) shift between the different charge storage states. The magnitude of the memory window will affect the accuracy of the data read.¹¹¹ Generally speaking, the larger the value, the higher the accuracy, but it will increase the energy consumption of the conversion between the stored energy states. The memory window value based on the PMMA dielectric layer is about 80 V.¹¹²

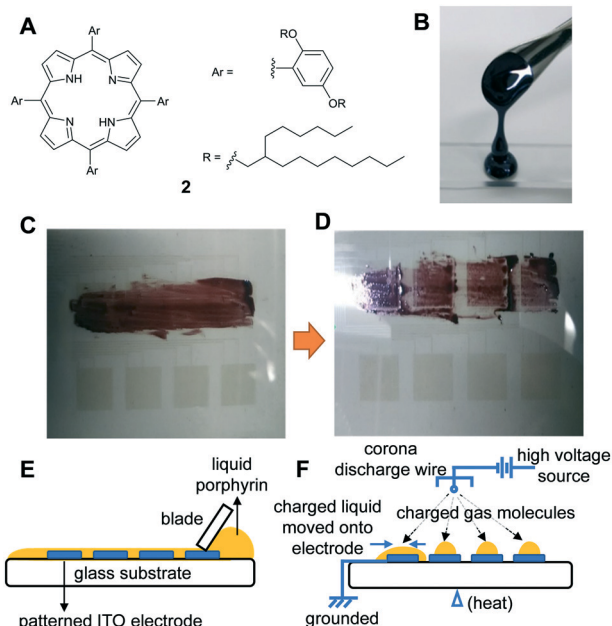


Fig. 6 (A) Chemical structure of alkyl-porphyrin liquid **2**. (B) Photograph of **2** under ambient light. Photographs of before (C) and after (D) corona charging of the liquid-porphyrin coated on patterned ITO glass electrode. (E) and (F) schematic diagram of the fabrication process of liquid-electret device on patterned ITO glass electrode. Reproduced from ref. 81 with permission from the ©2019 under a Creative Commons Attribution 4.0 Unported (CCBY) license.

conditions. Generally, FMLs are made up of uncharged molecular units and exist in amorphous states with a free-

flowing nature under ambient conditions. So far, most of the studies on FMLs^{17,84–88} have focused on their optical properties, such as luminescence.

The π -conjugated macrocycles are of interest for their applications in material chemistry and biological systems.^{89,90} The highly branched alkyl chains were attached on the periphery of porphyrin moiety (**2**, Fig. 6A) in order to shield the porphyrin core and provide the molecule with the nature of the liquid (Fig. 6B). We recently discovered porphyrin-based FMLs to be a “liquid electret”, where a well-shielded porphyrin core could stably hold an injected charge.⁸¹ The charged species could act as dielectric substances. Here, the liquid phase of the material is advantageous for storing the charge not only on the surface but also inside the material. Corona charging (Fig. 6C–F) was the method used to charge on **2**. The charged **2** was completely wetted on the patterned ITO surface and it spread out. The charged gas ions generated by the corona discharge flew to the surface of the film of **2** and were captured there. The charged liquid **2** moved itself onto the patterned ITO area by electrostatic force (Fig. 6E and F). This unique wettability change is one of the benefits of liquid electret device fabrication. Subsequently, the entire substrate was heated to further promote charge injection into the porphyrin core and a surface potential of -330 V was obtained. Then the charged liquid porphyrin electret **2** was placed between the ITO electrodes in such a way that the distance between the electrodes was $500\ \mu\text{m}$, which was managed by polyimide spacers (Fig. 7A). A voltage output of

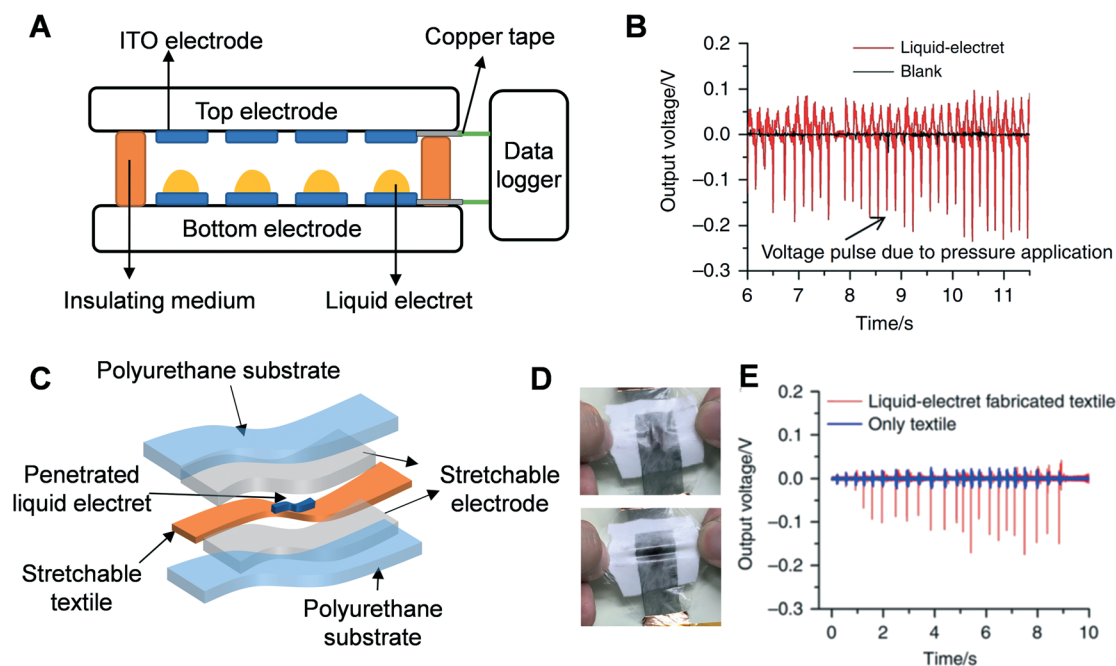


Fig. 7 (A) Schematics of cross-sectional view of liquid-electret device. (B) Output voltage characteristics of the electret device in its initial state. (C) Schematics of cross-sectional view of the stretchable liquid-electret device. (D) Image showing stretchable liquid-electret device prepared with a stretchable textile after stretching. (E) Comparison of output voltage characteristics of the stretchable liquid-electret device with textile with respect to presence and absence of liquid-electret. Reproduced from ref. 81 with permission from the ©2019 under a Creative Commons Attribution 4.0 Unported (CCBY) license.

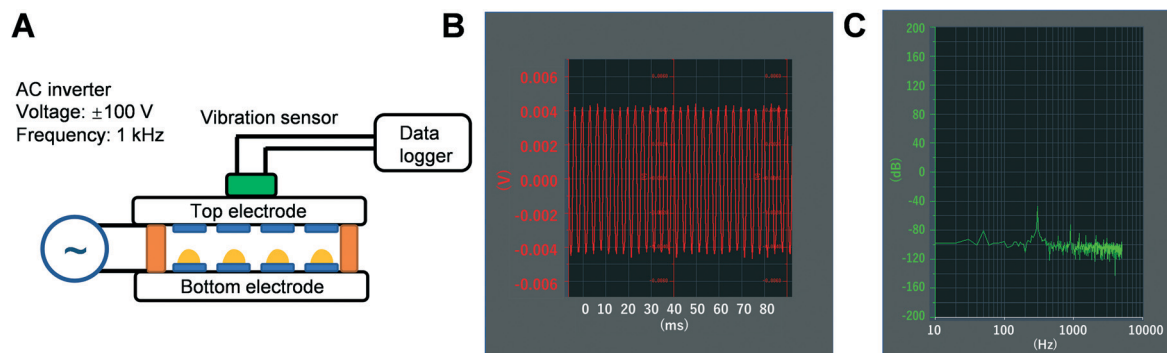


Fig. 8 (A) Schematic diagram of the cross-sectional view of the liquid-electret based electroacoustic device. (B) Output voltage from the vibration sensor. (C) Frequency characteristics of oscillation from the liquid electret device. Reproduced from ref. 81 with permission from the ©2019 under a Creative Commons Attribution 4.0 Unported (CCBY) license.

about -0.1 V was obtained when pressure was applied to the top electrode of the device (Fig. 7B). A stability study indicated a fast voltage decay during the first 12 hours and then the charge was stabilised for 43.5 days. The initial decay seems the process of neutralizing or flowing to the ground on the surface charge that is not trapped in the liquid electret. The properties of this liquid electret device are consistent with those of conventional solid film electret devices. The liquid characteristics of this electret allow it to be used for preparing stretchable devices. Polyurethane and silver-plated nylon fibres were used as stretchable materials, and the free-flowing property of **2** allowed it to easily penetrate the stretchable textiles (Fig. 7C and D). By applying finger pressure on this stretchable device, a voltage pulse of approximately -0.1 V was observed (Fig. 7E).

To further investigate the electroacoustic performance of the liquid electret, our group integrated the device with an alternating current (AC) inverter and a vibration sensor (Fig. 8A). When a 1 kHz input frequency was applied from the AC inverter, the oscillating frequency output in the electroacoustic device was approximately 200 Hz (Fig. 8B and C), which shows an audible sound.

Another molecular liquid-based electret is an ionic liquid-based electrostatic vibration energy collector with a 1 nm thick insulating gap, called an electrical double layer, reported by Sano *et al.*⁹¹ The insulating gap was generated at the interface with an ionic liquid filled between the electrodes, and an effective electrode distance of approximately $1\ \mu\text{m}$. By gently squeezing and relaxing the droplet by electrodes driven by a 1 Hz vibrator, an alternating current of 100 nA and a voltage of

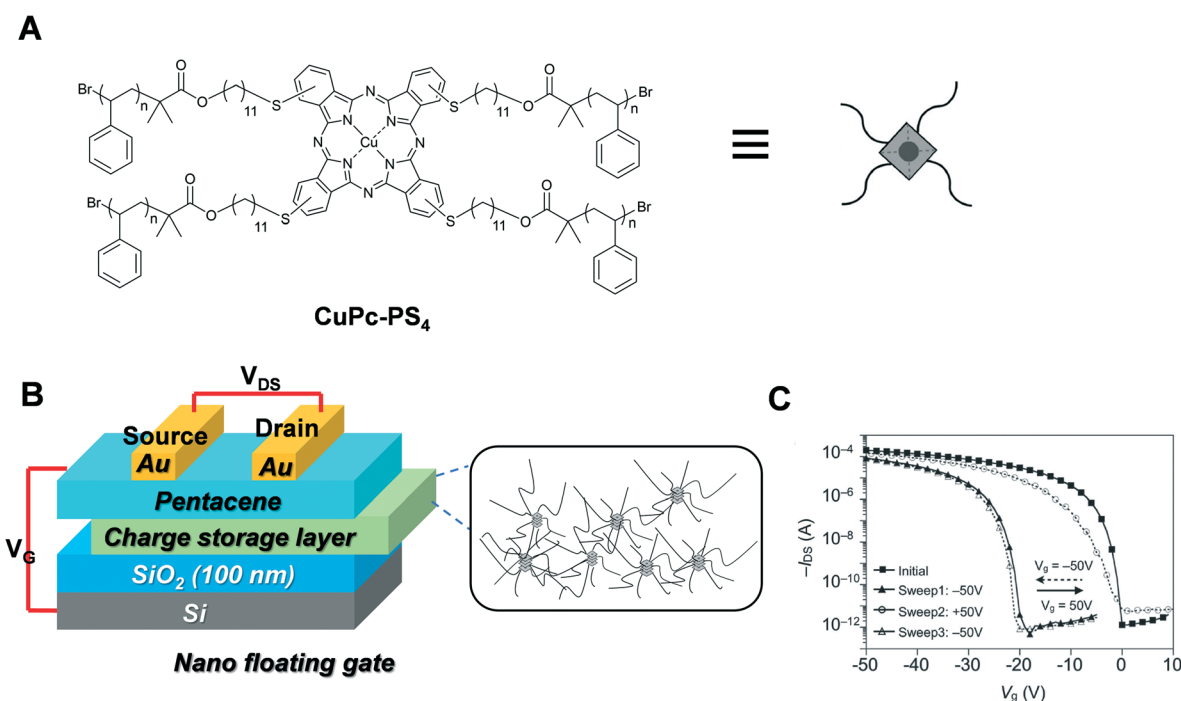


Fig. 9 (A) Chemical structure of CuPc-cored star-shaped polystyrene (CuPc-PS₄) and (B) schematic configuration of the OFET memory device with a nano floating gate. (C) Transfer characteristics of the pentacene OFET memory devices with CuPc-PS₄ charge storage layers. Adapted from ref. 92 with permission from the © 2015 the authors. Published by WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

100 mV were generated. This attempt has not been applied to stretchable electrets. However, with some ingenuity, fabrication of stretchable electret systems could be possible due to the highly deformable nature of the ionic liquid.

5. Applications of electrets

The electrets which are flexible and with longer charge life can be potentially used in various applications, including OFETs, energy harvesting devices, sensors, healthcare components and so on. In this section, the recently emerged applications of electret materials have been addressed.

5.1 Organic field-effect transistor memories

In the past few years, Aimi *et al.*⁹² developed a series of phthalocyanine (Pc) tethered block copolymers and applied them to a flash-type organic memory.⁸⁷ The organic layer material in the device was a four-arm star polystyrene (PS) embedded with a CuPc core (Fig. 9A). The authors found that the CuPc derivatives formed aggregates through π - π interactions. A study of the UV absorption spectrum showed that the CuPc core self-assembled in the polymer matrix to form aggregates even if there was a long-arm polymer present. Integrating such a polymer into an OFET memory device based on pentacene (Fig. 9B), it was found that the electrical output and transfer characteristics (Fig. 9C) differed greatly in comparison with a PS₄ memory device (without a CuPc core; data not shown), which also illustrated that its memory behaviour originated from the CuPc core. It is remarkable that the OFET device bearing a CuPc core as the charge storage layer provides a flash-type organic memory with a large 21 V memory window, a high $I_{\text{ON/OFF}}$ ratio of up to 10^7 , and long data retention characteristics.

Michinobu *et al.*⁹³ reported donor and acceptor functionalised copolymers as electret materials in preparing OFET devices. They designed the chemical frameworks as electret layers based on carbazole donors, thiophene spacers (3, Fig. 10), and the effect of an additional benzothiadiazole acceptor unit (4). The device properties show that the copolymer 3 provides effective and faster photoinduced recovery behaviour compared to the copolymer (4). The donor-acceptor-type conjugated polymers could facilitate the separation of the excitons (electron-hole pairs) by charge stabilisation. Thus, the generated excitons at the twisted donor-acceptor moieties could be efficiently dissociated into free charges. The electret based on donor-acceptor framework 4 can capture both the electron and hole, providing an ambipolar trapping behaviour. This is useful in multiple level data storage.

5.2 Energy harvesting devices

Energy harvesting is another important application of electrets. Due to the development of wearable flexible devices, materials with high biocompatibility are urgently needed and many researchers have recently shifted their focus to parylene. Because this is a highly adjustable polymer, the adaptability to complex shapes can be

combined with the high surface charge density. Here we highlight a recent report on an unusual fluorinated parylene for electret converter applications for energy harvesting.

Among the three common types of parylene (Table 1), the **parylene C** type has been the main focus because it is a polar material with piezoelectricity or ferroelectricity. Lagomarsini *et al.*³¹ studied the related electret properties (*e.g.* effect of charge polarity, material thickness, initial charging voltage, and temperature on the surface potential decay (SPD)). The authors found that the 9 μm thick **parylene AF-4** maintained a surface potential of -700 V for up to 280 days in the SPD test. Compared with **parylene C**, the non-polar nature of **parylene AF-4** can explain this long-term charge stability and its C-F bond position in the aliphatic chain, rather than the position of **parylene VT-4** in the aromatic chain. In addition, the 9 μm thick **parylene AF-4** exhibited a thermal stability of more than 100°C , and even at temperatures as high as 220°C , only a 50% reduction of the initial surface charge was observed. After 100 days, the surface charge density was still at a high level (positive charge: $1.79 \times 10^{-3} \text{ C m}^{-2}$; negative charge: $1.82 \times 10^{-3} \text{ C m}^{-2}$), which is comparable with **parylene C** (maximum surface charge density was 1.28 – $1.82 \times 10^{-3} \text{ C m}^{-2}$). Lo *et al.*²⁹ developed an electret based on **parylene AF-4** thin film fabricated by chemical vapor deposition (CVD) method. It was investigated in terms of process, stability, temperature reliability and optimisation. The charge density is $3.7 \times 10^{-3} \text{ C m}^{-2}$, which is the highest value among polymer electrets reported so far.^{9,31,35,52,55}

5.3 Electret sensors and actuators

A key component of various detectors or monitoring systems is a sensor, which includes the information collection part to the transmission part, back-end storage, control, and display equipment. Acoustic-electric sensors (*e.g.* microphones^{94–97}), electro-acoustic sensors (*e.g.* earphones^{97,98} and speakers⁹⁶) and mechanical force-electricity sensors (*e.g.* pickups^{96,99}) are typical examples. For human wearable device applications, scientists have mainly focused on visual and auditory research, with relatively few haptic sensors. Recently, Zhong *et al.*¹⁰⁰ developed a flexible thin-film piezoelectric electret actuator/sensor that can be used for human-machine interfaces. It is made of FEP electret films on the top and bottom sides, the corresponding gold and aluminium electrodes, and an Ecoflex§ spacer in the middle. This is a flexible sandwich structure with a total thickness of $150 \mu\text{m}$, and the equivalent piezoelectric constant $d_{33}\P$ is as high as

§ Ecoflex: Ecoflex is an aliphatic-aromatic random copolyester manufactured by BASF in Germany. It is made from adipic acid, terephthalic acid, and 1,4-butanediol.

\P d_{33} : d_{33} is one of the most commonly used important parameters for characterising the performance of piezoelectric materials. It is used to measure the strength of the piezoelectric effect. Generally, the higher the piezoelectric constant, the better the piezoelectric performance. The first number in the subscript refers to the direction of the electric field and the second number refers to the direction of the stress or strain; “33” means that the polarisation direction is the same as the force applied during the measurement.

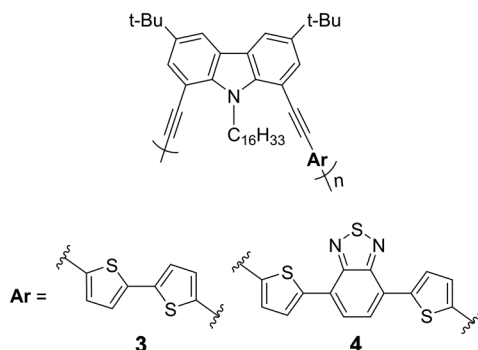


Fig. 10 Chemical structure of carbazole/bithiophene copolymer **3** and carbazole/thiophene-benzothiadiazole-thiophene copolymer **4**.⁹³

$4.05 \times 10^{-9} \text{ C N}^{-1}$, which is higher than many traditional piezoelectric materials such as PZT ($d_{33} = 2.0\text{--}6.0 \times 10^{-10} \text{ C N}^{-1}$)^{101–104} and piezoelectric electret materials such as cellular PP.¹⁰⁵ On the thin-film array, each work unit can be switched between the functions of the sensor and the actuator at will. In addition, the human body signal measured by the sensor can be used to control the behaviour of the actuator, realising mutual cooperation in function. In the actuator mode, the mechanical vibration of a single actuator can generate a driving force of more than 0.02 N, which can be easily sensed by human skin. In the sensor mode, a single sensor can detect a pressure as small as 1.84 Pa, and the output signal changes less than 1% in more than 6000 cycles of loading. This sensor unit can be used to measure the physiological state of the skin surface including the pulse and muscle movement.

6. Summary and perspectives

In this mini review, we first briefly introduced the basic information and types of electrets, and then we discussed common charging methods including corona charging, electron beam irradiation charging, liquid contact charging, and electrothermal polarisation charging. We also focused on the development and applications of organic soft electret materials. We summarised a total of five approaches to achieve soft/stretchable electrets: (i) introduction of pores in hard electrets, (ii) composites with carbon nanotubes or nanoparticles, (iii) preparation of flexible thin sheets, (iv) use of block copolymers, and (v) functional molecular liquids. If the flexible/stretchable electrets are further developed and considered in more realistic applications, a number of problems remain to be solved. The following is a summary of the major challenges.

a. Promoting flexibility

The described methods for making electrets “flexible” are still in the initial stage of development, and further attempts are required. Based on the “alkyl- π engineering strategy”,¹⁰⁶ several fluidic materials developed by our group in recent years have substantially assisted the

research of flexible/stretchable applications. Similarly to alkylated porphyrin liquid (**2**), conjugated polymers can be greatly softened by introducing branch alkyl side chains, which can act as an “internal plasticizer”.^{107–110} The viscoelasticity of conjugated polymer fluids can be easily adjusted by introducing a bulky or branched alkyl side chains (different alkyl chain length).¹⁰⁹ This softening methodology for conjugated polymers could also be applicable for the development of flexible/stretchable electrets.

b. Simplifying charging method

Although we have outlined several means for electret charging in this review, the technology is not mature yet and is only suitable for a small number of experiments. The charging methods developed so far are not sufficient for large-scale (industrial-scale) electret production. It is worth mentioning that if the electrothermal polarisation charging method can solve the issue of uneven lateral distribution of surface and body charges, it will be the first choice for large-scale charging, particularly for the polarisation of piezoelectric electret materials.

c. Increasing charge density

Charge density is an important parameter used to evaluate the charge storage capacity of electrets, and it includes the surface charge density and space charge density. So far, most electret charges are stored on the surface of dielectric materials, which is insufficient for practical applications. If the charge can be generated inside the dielectric material, this will increase the total charge density. The scientific community is making efforts in this direction, such as by creating pores in hard materials and manufacturing liquid electrets, but more progress with sufficient charge density is still required. AI-assisted machine learning would be a promising approach for predicting the high charge density in electret molecules or polymers.⁵⁶

d. Improving charge lifetime

In general, organic electret materials tend to have a longer charge life than inorganic electrets. However, electrets made of stretchable dielectric materials often exhibit a short charge life. The lifetime of the electret also depends on the type of sealing material, the method of formation, environmental factors, and so on. How to further increase the charge life under the premise of achieving flexible/stretchable electrets and retaining high charge density will be a huge challenge in the future.

If the major challenges mentioned above can be overcome, flexible/stretchable electrets could enjoy unprecedented development in the next few decades, leading to more applications in soft electronic devices.

Author contributions

Mr. Z. Guo and Dr. Y. Patil contributed equally to the preparation of this mini-review. They collected and analysed the related literature and wrote the manuscript draft. Dr. A. Shinohara, Dr. K. Nagura, and Dr. M. Yoshida participated in the analysis of the literature and polishing the draft. Prof. T. Nakanishi is the main creator of the principle for this work and a corresponding author. He also guided the writing and polishing of the draft. All authors have read and agreed to the final text.

Conflicts of interest

There are no conflicts to declare.

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